TECHNICAL MANUAL

My Health Passport

System Architecture, SLK-581 & SHA-256 Technical Reference, and Researcher Integration Guide

Document version: 2.0

Prepared for: Research teams integrating with My Health Passport

Applies to: index.html, portal.html, phone.html, link_gen.html

Contents

1. Executive Summary

What this document covers, and who it is for

> **My Health Passport** is a small set of static, client-side web pages that let a research participant generate a pseudo-anonymised identifier — a **My Health Passport Number** (an SLK-581 statistical linkage key[1]) or, optionally, a **My Health Passport Barcode** (a SHA-256 cryptographic hash[2] of that number) — entirely inside their own browser, and share only that identifier with a research team.
>
> This manual is written for researchers, data managers, and HREC applicants who need to integrate My Health Passport into a REDCap or Qualtrics survey, a phone-based interview workflow, or an email-based study contact process. It supersedes all earlier revisions of this guide.

The system is deliberately simple: there is no server, no database, and no account system. Every calculation — building the SLK-581, hashing it, checking a form field — happens locally on the participant's device. The pages exist only to standardise how that identifier is generated and handed off to a survey platform, a phone, or an email client.

What has changed since the previous edition of this guide:

- The main generator page has been renamed and now sits at the root of the project as `index.html` (previously distributed as `SLK581_builder.html`).
- A dedicated **Portal Link Generator** (`link_gen.html`) is now available, removing the need to hand-build query strings for most researchers.
- `phone.html` now supports both an SMS reply workflow and a new **email reply workflow**.
- This edition adds a full technical briefing on SLK-581 and SHA-256, with cited references, for teams that need to justify the approach in an ethics submission or data governance review.

2. System Architecture & File Structure

Where everything lives, and what each file does

My Health Passport is a static site: four HTML files sitting in a single root directory, with no build step and no server-side component. All four files are self-contained — each can be opened directly in a browser or hosted from any plain web server or static host (e.g. GitHub Pages).

File	Role	Typical audience
`index.html`	The main public page. Explains the project, generates a Passport Number/Barcode directly, and links to the Technology, Privacy, and Resources sections.	Participants arriving directly; general public
`portal.html`	A focused landing page for survey-linked participants. Generates the identifier and, on consent, redirects to a survey URL with the identifier appended as a query parameter.	Participants following a REDCap/Qualtrics invitation link
`phone.html`	A landing page for phone- or email-based studies. Generates the identifier and, on consent, hands off to the participant's SMS or email app with the identifier pre-filled in a reply.	Participants contacted by phone interview or email
`link_gen.html`	An internal tool for researchers. Builds correctly-formatted `portal.html` / `phone.html` links (with QR code output) without needing to hand-construct query strings.	Researchers, study coordinators

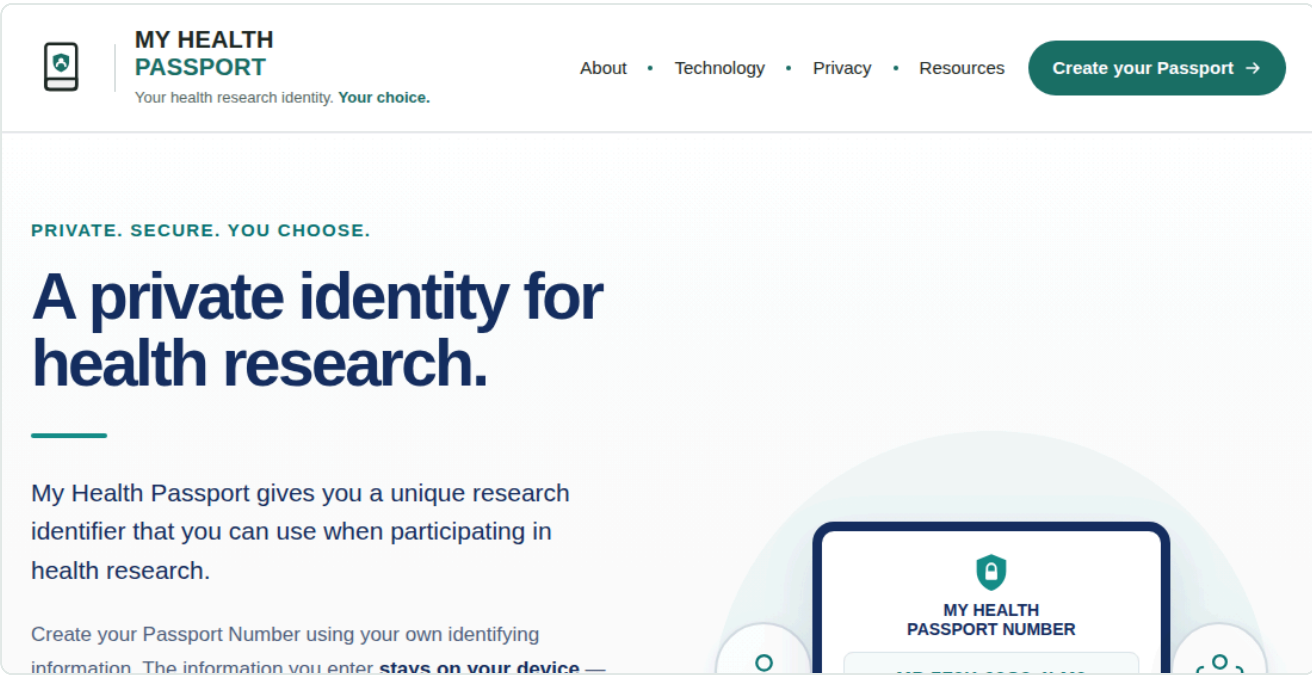

Figure 1 — `index.html`, the project's public entry point, showing the shared header and navigation used across all four pages.

Researcher opens `link_gen.html` → Builds a link to `portal.html` or `phone.html` → Participant opens the link, generates their identifier → Identifier reaches the survey, phone, or inbox

Deployment note: because all four files live in the same directory, links between them are simple relative paths (e.g. `portal.html`, `phone.html`). If this project is deployed under a different base path or domain than shown in this manual's examples, adjust the base URL only — every query parameter described in this document is unaffected by where the files are hosted.

3. Technical Briefing: The SLK-581 Statistical Linkage Key

What it is, how it is built, and what it can and cannot do

3.1 Origin and purpose

The Statistical Linkage Key, 581-character format (SLK-581) is a data linkage standard developed and published by the Australian Institute of Health and Welfare (AIHW)[3]. It was created to solve a specific problem in health and community-service data collection: allowing records relating to the same person to be brought together across separate datasets, without those datasets ever exchanging a person's name, address, or other directly identifying information.

Rather than encrypting or storing a name, SLK-581 derives a short, largely non-reversible code from a small number of demographic elements the person already knows: parts of their family and given name, their date of birth, and their recorded sex. Two records created from the same underlying details will produce the same SLK-581, which is what allows linkage — without either dataset needing to see the other's raw identifying fields.

3.2 Construction algorithm

My Health Passport implements the SLK-581 algorithm exactly as specified by AIHW[4]. The 14-character key is assembled from four components, concatenated in order:

Component	Characters	Rule
Family name	3	The 2nd, 3rd and 5th letters of the family name, counting alphabetic characters only. A missing letter (e.g. a short surname) is filled with the digit `2`.
Given name	2	The 2nd and 3rd letters of the first given name, with the same fallback rule.
Date of birth	8	Recorded as `DDMMYYYY`.
Sex	1	A single digit: 1 (male), 2 (female), 3 (intersex/indeterminate), or 9 (not stated), consistent with the ABS/AIHW standard for recording sex.

Worked example

Family name `O'Brien-Smith` → alphabetic letters `OBRIENSMITH` → 2nd/3rd/5th letters → `BRE`
Given name `Elizabeth` → 2nd/3rd letters → `LI`
Date of birth 15 June 1990 → `15061990`
Sex: Female → `2`

Resulting SLK-581: `BRELI150619902`

This is exactly the calculation performed by the generator on `index.html`, `portal.html`, and `phone.html` — entirely in the participant's browser, using JavaScript equivalent to the reference implementation above.

3.3 Properties and limitations

- **It is not encryption.** SLK-581 is a deterministic transformation, not a cryptographic scheme. Given a plausible name, date of birth, and sex, the same key can in principle be reconstructed — SLK-581's privacy value comes from what it withholds (the full name, address, and other fields), not from mathematical irreversibility.
- **It is not guaranteed unique.** Two different people can share an SLK-581, particularly common names combined with common birth dates. Independent AIHW evaluation of SLK-based linkage against name-based linkage found strong but not perfect performance: a stepwise SLK-based strategy achieved a 99.7% positive predictive value and 98.5% sensitivity against a name-based reference standard in one aged-care linkage study[5], and AIHW has separately reported that a small proportion of residents in large administrative datasets share an SLK-581 with at least one other person[6].
- **It is widely used, not experimental.** SLK-581 underpins linkage in numerous Australian health and community-service data collections and has been the subject of peer-reviewed evaluation in registry-based research[7].
- **Governance still matters.** AIHW and equivalent bodies recommend SLK-581 be used within an appropriately governed linkage project — with ethics approval, and additional identifying variables where higher linkage confidence is required — rather than as a stand-alone guarantee of identity.

> **Do not present SLK-581 as anonymisation.** It is more accurately described as *pseudonymisation*: identifying detail is withheld from the receiving dataset, but the key is derived from, and in some circumstances could be linked back to, identifying information the participant already knows. HREC submissions should describe it in these terms — see Section 12.

4. Technical Briefing: The SHA-256 Cryptographic Hash

The optional "My Health Passport Barcode"

4.1 What SHA-256 is

SHA-256 ("Secure Hash Algorithm, 256-bit") is a member of the SHA-2 family of cryptographic hash functions, specified by the U.S. National Institute of Standards and Technology (NIST) in Federal Information Processing Standard 180-4[8]. A hash function takes an input of any length and produces a fixed-length output — for SHA-256, always 256 bits, conventionally written as 64 hexadecimal characters.

Three properties matter for this project:

- **Deterministic:** the same input always produces the same 64-character output, which is what allows the same Passport Number to always generate the same Barcode.
- **One-way:** there is no practical method to recover the original input from the hash output alone. This is a fundamentally different property to SLK-581, which withholds detail but does not cryptographically destroy it.
- **Avalanche effect:** changing a single character of the input (e.g. one letter of a name) produces a completely different, unrelated-looking 64-character output, which is why a Passport Barcode gives no partial information about the Passport Number it came from.

4.2 Why it is used here (the "My Health Passport Barcode")

When barcode mode is enabled, the portal computes `SHA-256(Passport Number)` locally in the participant's browser, using the standard Web Cryptography API (`crypto.subtle.digest`), and displays and shares that hash — the **My Health Passport Barcode** — instead of the raw SLK-581. This gives studies a way to obtain a stable, unique-per-input research identifier while adding a one-way transformation on top of SLK-581, for projects that want that additional layer.

SLK-581 generated locally → **SHA-256 hashed locally (browser)** → **64-character Barcode shared with researcher**

Because the hash is one-way, reversing it to recover the original SLK-581 — and therefore the demographic detail behind it — is not practically feasible. It is important, however, to describe this accurately rather than in absolute terms: cryptographic best practice is to describe SHA-256 as making reversal *computationally infeasible* with current methods, not mathematically impossible for all time.

4.3 Standards and approvals

SHA-256 is standardised by NIST under FIPS 180-4[8] and is one of the hashing algorithms approved for general-purpose use by the Australian Signals Directorate (ASD) under the Australian Government

Information Security Manual (ISM)[9]. Current ASD guidance approves SHA-2 family algorithms (including SHA-256) for general use, while recommending that new systems intended for long-term use prefer the longer SHA-384 or SHA-512 variants, noting that ASD's approval of SHA-224 and SHA-256 specifically is not intended to extend beyond 2030 for interoperability planning reasons. This is a reasonable, currently-approved choice for this project's purpose; it is not a claim that SHA-256 is invulnerable indefinitely.

5. Portal Workflow Overview

How a participant experiences the process end to end

1. **Participant landing:** the participant opens a link to `index.html`, `portal.html`, or `phone.html`, depending on how they were contacted.
2. **Identity key generation:** the participant enters their family name, given name, date of birth, and sex. The page builds their SLK-581 (and, if enabled, its SHA-256 Barcode) live, showing a progress indicator as each field is completed.
3. **Consent to share:** the participant reviews the generated identifier and taps the clearly labelled share action — "Share your My Health Passport Number/Barcode" (survey), or the equivalent SMS/Email reply action.
4. **Hand-off:** depending on the page, this either opens the destination survey in a new tab with the identifier appended as a URL parameter, or opens the participant's own SMS or email app with a pre-filled reply.

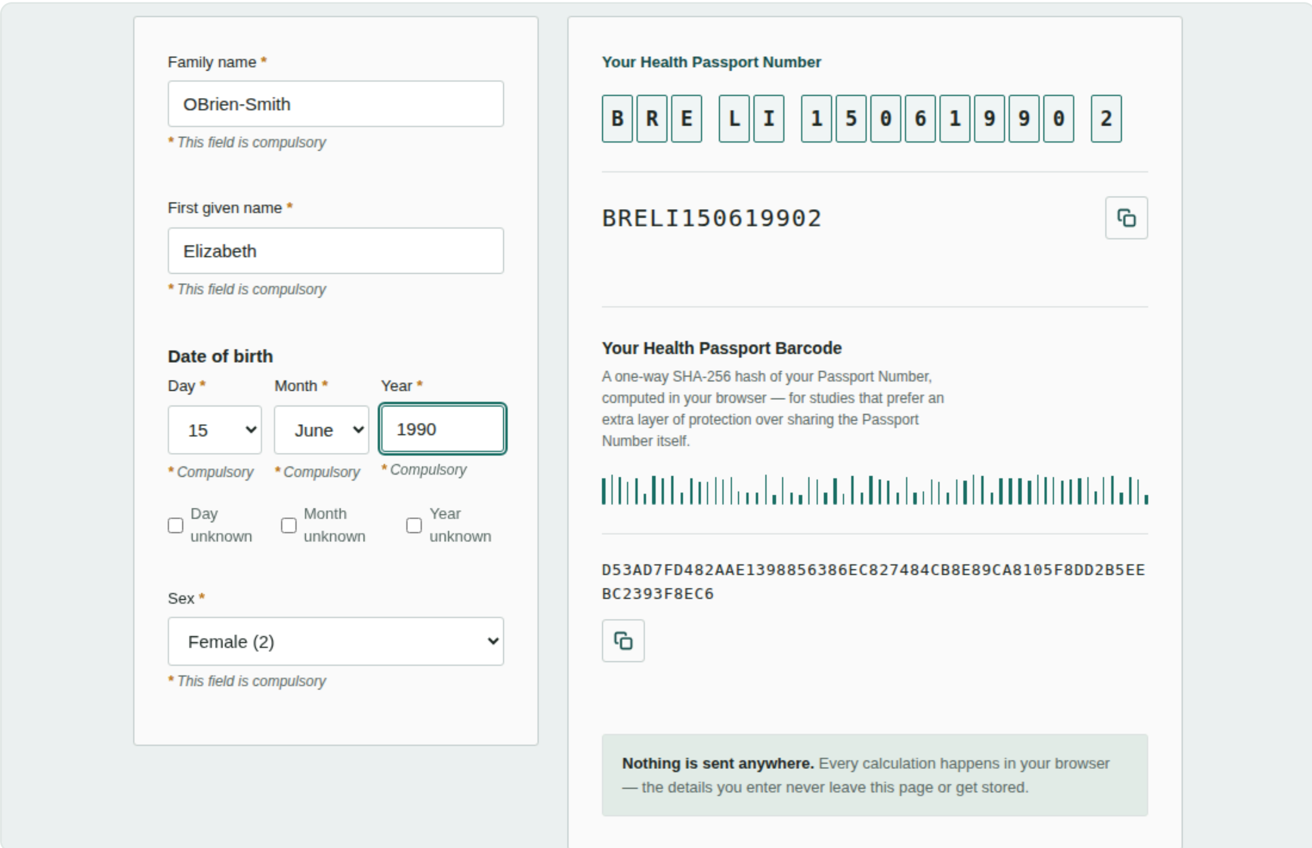

Figure 2 — The generator on `index.html`, showing a completed Passport Number and its optional SHA-256 Barcode.

At no point does any of the entered detail (name, date of birth, sex) leave the participant's device. Only the finished identifier — the SLK-581 or its SHA-256 Barcode — is ever transmitted, and only once the participant actively chooses to share it.

`portal.html` presents the same generator inside a three-step flow, with each step lighting up as the participant progresses:

MY HEALTH
PASSPORT PORTAL

Unlock Your Survey

1 Enter Your Details

Fill in each field below. Nothing you enter here is sent anywhere — only the finished Passport Number is ever shared.

Family name *

OBrien-Smith

**This field is compulsory*

First given name *

Elizabeth

**This field is compulsory*

Date of birth *

15 | June | 1990

Sex *

Female

2 Get your Unique ID

Your Anonymised My Health Passport Number

BRELI150619902

3 Share your My Health Passport Number

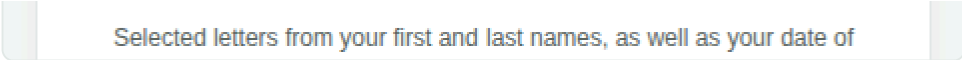

Figure 3 — `portal.html` 's step-by-step flow: enter details, generate the ID, then share it with the survey.

6. Using the Portal Link Generator (link_gen.html)

The recommended way to build a distribution link

Rather than hand-constructing query strings (Section 7), most researchers should use `link_gen.html`. It asks three questions and builds the correct link automatically:

1. **Link type** — Online Survey, SMS phone message, or Email.
2. **Contact detail** — the survey name and URL, the reply phone number, or the reply email address, depending on the link type chosen.
3. **Identifier type** — My Health Passport Number (SLK-581) or My Health Passport Barcode (SHA-256).

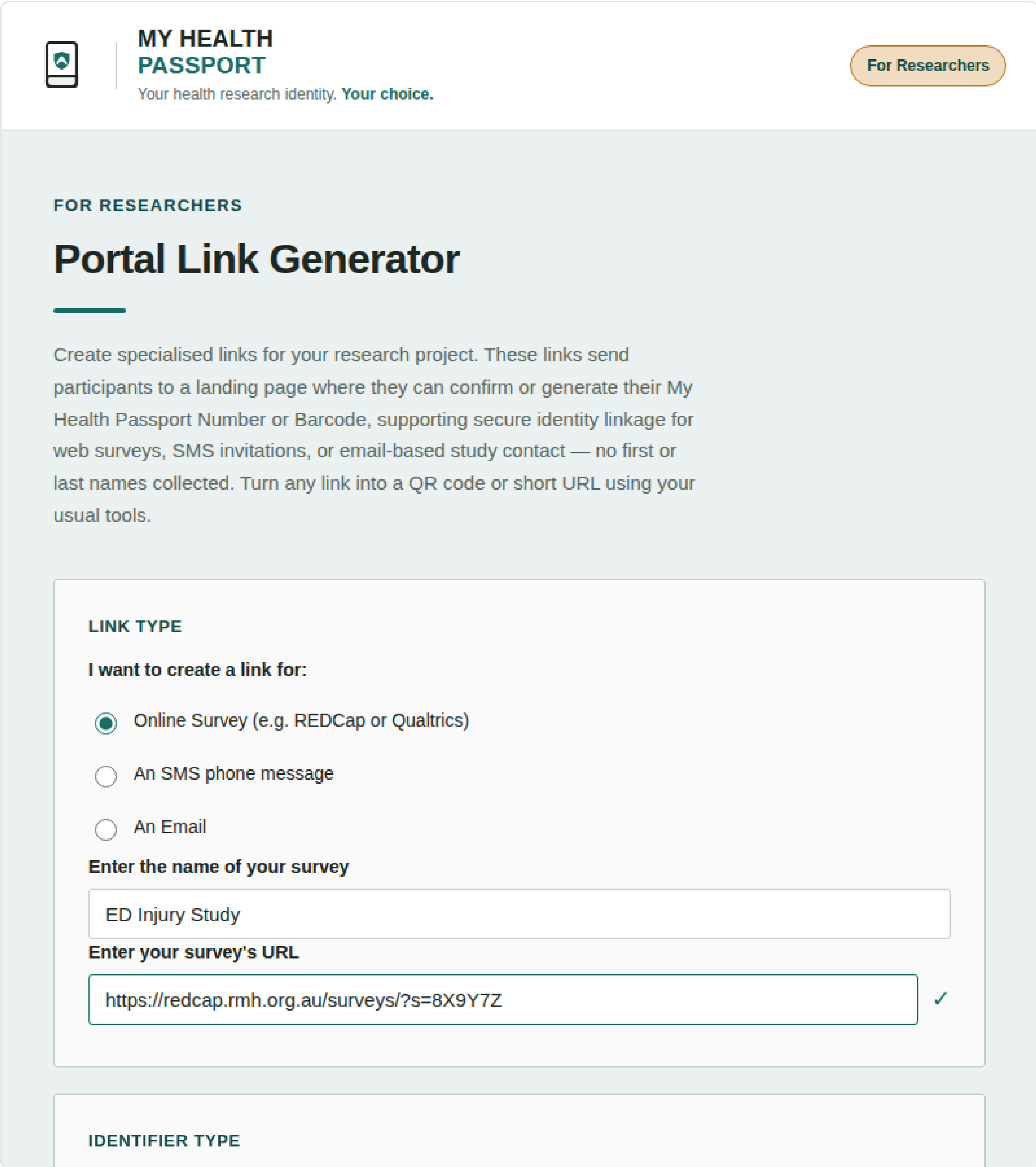

Figure 4 — `link_gen.html`, configured to build a survey link for an "ED Injury Study".

The tool live-validates a survey URL (it must begin with `https://`), generates the finished link in a copyable field, and can render that link as a scannable QR code for printed materials — useful for posters, flyers, or waiting-room signage.

> **Where the QR code comes from:** the QR code image is produced by a third-party rendering service, which receives only the finished portal link itself (never any participant data, since no participant has interacted with the link yet). Study teams with strict data-egress requirements should be aware that this one step involves an external request, unlike every other part of this system.

7. Constructing Links Manually (Advanced)

For teams integrating link generation into their own tooling

Both `portal.html` and `phone.html` read plain URL query parameters. This section documents them directly, for teams who want to generate links programmatically (e.g. from a REDCap API or a mail-merge) rather than via `link_gen.html`.

portal.html (survey links)

```
portal.html?siteTitle=[SURVEY_NAME]&siteUrl=[SURVEY_URL]&barcode=[true|false]
```

Parameter	Required	Notes
`siteTitle`	Yes	Shown in the Share button label. Spaces must be URL-encoded as `%20`.
`siteUrl`	Yes	The destination survey URL. The identifier is appended to this as `?slk=` or `&barcode=`.
`barcode`	No (default `false`)	`true` shares the SHA-256 Barcode instead of the raw SLK-581.

phone.html (SMS and email links)

```
phone.html?SurveyPhone=[PHONE]&barcode=[true|false]&linkType=sms
phone.html?replyemail=[EMAIL]&barcode=[true|false]&linkType=email
```

Parameter	Required	Notes
`SurveyPhone`	For SMS links	The number the participant's reply SMS should be addressed to.
`replyemail`	For email links	The address the participant's reply email should be addressed to.
`linkType`	Recommended	`sms` or `email`. Tells the page which reply channel to present. If omitted, the page infers the channel from whichever of `SurveyPhone` / `replyemail` is present.
`barcode`	No (default `false`)	As above — when `true`, the SMS or email body contains the SHA-256 Barcode instead of the raw SLK-581.

Example — REDCap survey, Barcode mode enabled

`portal.html?siteTitle=ED%20Injury%20Study&siteUrl=https://redcap.rmh.org.au/surveys/?s=8X9Y7Z&barcode=true`

Example — phone interview, SMS reply

`phone.html?SurveyPhone=%2B61400000000&barcode=false&linkType=sms`

Example — email-contacted participant

`phone.html?replyemail=study%40example.org&barcode=false&linkType=email`

8. Survey Platform Setup (REDCap & Qualtrics)

Capturing the identifier once it reaches your survey

> **Critical requirement for multi-page surveys:** the receiving field **must be on the very first page** of the survey. If it sits on page 2 or later, page navigation strips the incoming URL parameter before it can be captured.

REDCap setup

1. Open the project's **Online Designer**.
2. On the **first survey instrument**, add a new Text Box field.
3. Set the Field Name to exactly `slk` — or `barcode` if the link uses Barcode mode.
4. Under Action Tags, add `@READONLY` to prevent participant editing.

Qualtrics setup

1. Open the **Survey Flow** tab.
2. Add a new **Embedded Data** element.
3. Name the field exactly `slk` — or `barcode` if the link uses Barcode mode — and leave the value blank.
4. Drag the Embedded Data block to the very top of the Survey Flow, above Question Block 1.

9. SHA-256 Encrypted Barcode Mode

Enabling and receiving the hashed variant

Barcode mode (`barcode=true`) is described technically in Section 4. Operationally, enabling it changes three things simultaneously, on both `portal.html` and `phone.html` :

- The participant sees the "My Health Passport Barcode" panel instead of the raw Passport Number panel.
- The Share button label and consent disclaimer update to describe sharing a Barcode rather than a Passport Number.
- The value transmitted — whether by URL parameter, SMS, or email — is the 64-character SHA-256 hash, never the underlying SLK-581.

MY HEALTH
PASSPORT PORTAL

Generate your secure Phone ID

1 Enter Your Details

Fill in each field below. Nothing you enter here is sent anywhere — only the finished Passport Number is ever shared.

Family name *

OBrien-Smith

**This field is compulsory*

First given name *

Elizabeth

**This field is compulsory*

Date of birth *

15 | June | 1990

Sex *

Female

2 Get your Unique ID

Your Anonymised My Health Passport Barcode

D53AD7FD482AAE1398856386EC827484CB8E89CA81
05F8DD2B5EEBC2393F8EC6

Reading your ID aloud

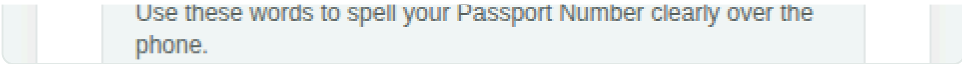

Figure 5 — `phone.html` in Barcode mode: the SLK-581 panel is replaced entirely by the SHA-256 Barcode panel.

> **Field name differs from standard mode.** When a link uses Barcode mode, the receiving field on the survey side must be named `barcode`, not `slk` (see Section 8). The two modes are mutually exclusive per link — build and label only the field matching the mode your distribution link actually uses.

10. Mobile Phone / SMS Reply Workflow

For phone interviews and other non-web-survey studies

For phone-based interviews, `phone.html` lets a participant generate their identifier on their own device and reply directly by SMS — useful for establishing an anonymous, persistent identifier at the very start of an interview, before any interview content is discussed.

1. **Send the link:** at the start of the call, the interviewer sends the `phone.html` link (built via `link_gen.html`, see Section 6) to the participant by SMS.
2. **Participant generates their ID:** the participant taps the link and completes the form on their own device.
3. **Participant replies:** tapping the Share action opens the participant's own messaging app, pre-filled with a message addressed to the study's reply number, containing the generated identifier.
4. **Researcher logs the ID:** once sent, the interviewer receives the SMS and records the identifier against the interview record.

Verbal fallback: `phone.html` also renders each character of the Passport Number using the NATO phonetic alphabet (e.g. `B` → "Bravo"), so an interviewer can capture it verbally during the call if a written record is needed immediately, without waiting for the SMS to arrive.

> **Delivery is participant-initiated.** The Share action opens a pre-filled message in the participant's own app — it does not send automatically. The participant must still tap Send. Interviewers should confirm receipt before proceeding, particularly in time-sensitive settings.

11. Email Reply Workflow

New: for participants contacted by email

`phone.html` also supports an email-based reply, for studies that make first contact by email rather than phone. The mechanism mirrors the SMS workflow in Section 10, substituting a `mailto:` hand-off for an `sms:` hand-off.

1. Build an email-type link in `link_gen.html`, supplying the study's reply email address.
2. Send that link to the participant by email, as part of the standard study invitation.
3. The participant opens the link, completes the form, and taps **"Reply via Email with your My Health Passport Number"** (or Barcode, if enabled).
4. Their default mail client opens with a new message addressed to the study's reply address, subject and body pre-filled with the identifier. The participant reviews and sends it.

Which reply channel is shown is controlled by the `linkType` parameter (`sms` or `email`) described in Section 7; the page falls back to inferring the channel from whichever of `SurveyPhone` or `replyemail` is present if `linkType` is omitted, for compatibility with earlier links.

> **Consistent with the rest of the system:** as with SMS, no identifying detail is transmitted automatically — only the finished identifier, and only once the participant reviews and sends the pre-filled email themselves.

12. HREC Ethics Statement Template

Suggested wording for ethics submissions

Researchers may adapt the following text for Human Research Ethics Committee (HREC) submissions. It has been updated to reflect the email reply channel and to describe SLK-581 as pseudonymisation rather than anonymisation, consistent with Section 3.3.

> *"Data collection will utilise the My Health Passport tools as a secure participant landing page. Participants will generate their Statistical Linkage Key (SLK-581) on their own device; no first or last names, addresses, or other directly identifying fields are captured or transmitted at any point. Upon explicit participant authorisation, only the resulting pseudonymised SLK-581 code — or, where the study elects to use the optional SHA-256 hashed Barcode mode, a one-way cryptographic hash derived from it — is transmitted into the research record, via URL parameter injection into the survey platform, or via a participant-initiated SMS or email reply. SLK-581 pseudonymises rather than anonymises participant data: while no directly identifying fields are transmitted, the key is derived from identifying information the participant already holds, and record linkage carries a small residual risk of collision between participants with similar names and birth dates, as documented in published AIHW evaluations."*

Appendix A — File & URL Quick Reference

File	Purpose	Key parameters
`index.html`	Main public page and stand-alone generator	—
`portal.html`	Survey-linked participant landing page	`siteTitle` , `siteUrl` , `barcode`
`phone.html`	Phone/email-linked participant landing page	`SurveyPhone` , `replyemail` , `linkType` , `barcode`
`link_gen.html`	Researcher tool for building the above links, with QR output	—

Survey field name	When to use it
`slk`	Standard mode (raw SLK-581 Passport Number)
`barcode`	Barcode mode (SHA-256 hash of the Passport Number)

References

Cited works, in order of first appearance

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